

Don't Trust the Experts!

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1 Background

Good's "value of information" theorem: if

- You have probabilistically coherent credences
- You are certain that your future credences will be derived from your current credences by conditionalizing on whichever is the true member of some partition of propositions.
- You are certain that in future you will handle any decisions by picking some option that maximizes the expectation of a certain function u , which is in fact your current utility function.

then in a decision with a bunch of ordinary options and one special option in which you delay the decision among the ordinary options to the future, you will never prefer any of the ordinary options to the special option.

Good's philosophical interpretation: rational people should always gather information before making a decision, so long as it is available for free.

There is something rather attractive about this conclusion. Of course there are special cases like that of Ulysses and the sirens where it's wise to avoid leaving decisions to one's future self. But such cases seem to turn on (possibilities of) pathological goings-on, which can plausibly often be harmlessly ignored.

But the idea that (even in normal cases) our future credences come from our current credences by conditionalizing on whichever is the true member of some partition is very implausible (except as an approximation), for several reasons.

- That would involve having credence 1 in lots of contingent truths. Given the standard behavioral profile of credence 1, this seems extreme.
- It would involve a substantial chance of 'cliff-edge' credence 1: having credence 1 in a proposition that is almost false, and could very easily have been false.

- There seems to be no way in which one could be *reliably* set up to always conditionalize on whichever is the *true* member of a nontrivial partition: near the cell-borders, there'll be a substantial risk of conditionalizing on the wrong (false) cell. It doesn't seem reasonable to have dispositions involving such risks.

2 Terminology

Fix a set W (of "worlds") and an algebra $\mathcal{F}$ over W .

- A **probability-function-valued random variable**, or for short a **frame**, is a function $\mathcal{P}$ that maps every world to a probability function on $\langle W, \mathcal{F} \rangle$.
- An **option** O is a function from worlds to real numbers (intuitively: the utility payoff of the option in the world)
- A **decision problem** D is a finite set of options
- A **strategy** S for a decision problem D is a function that maps each world to one of the options in D .
- Strategy S is **allowed** by frame $\mathcal{P}$ iff for every world w , the expected value of $S(w)$ in $\mathcal{P}(w)$ is at least as high as that of any element of D .
- The **payoff** of strategy S is the function that maps each world w to $S(w)(w)$ (the value at w of the option that S selects at w)
- The **expected value** of a strategy S in a probability function π is just the expected value in π of S 's payoff.
- Then, probability function π **Values** frame $\mathcal{P}$ iff for every decision problem D , the expected value in π of any strategy for D allowed by $\mathcal{P}(w)$ is at least as high as the expected value in π of any of the options in D .
- Given a probability function π and a partition $\{p_i\}$ of propositions with positive π -probability, $\mathcal{P}_{\pi, \{p_i\}}$ is the frame whose value at a world w is $\pi(\cdot \mid p_j)$, where p_j is whichever member of the partition is true at w .

Then what Good's theorem says is: for any π and $\{p_i\}$, π Values $\mathcal{P}_{\pi, \{p_i\}}$.

Question: if π Values $\mathcal{P}$, does that mean that there is a partition $\{p_i\}$ such that $\mathcal{P} = \mathcal{P}_{\pi, \{p_i\}}$?

Answer (Dorst): no.

Counterexample: a 2-world frame with

$$\begin{array}{lll} \pi(\{w_1\}) = 0.5 & \mathcal{P}(w_1)(\{w_1\}) = 0.6 & \mathcal{P}(w_2)(\{w_1\}) = 0.4 \\ \pi(\{w_2\}) = 0.5 & \mathcal{P}(w_2)(\{w_1\}) = 0.4 & \mathcal{P}(w_2)(\{w_2\}) = 0.6 \end{array}$$

3 Value

This suggests the interesting possibility of accepting Good's conclusion that we should Value our future credences (pathologies aside) without accepting his premise (that these arise by conditionalization on a partition).

But how do we set the pathologies aside? One approach (especially common at MIT) is to appeal to a highly idealized notion of rationality on which for each agent x and time t , there's exactly one probability function $Cr_{x,t}$ such that x *should* (rationally) have it as their credence function at t . For any x, t , let the **should-frame** for x and t be the frame whose value at each world is the probability function that should be x 's credence at t at that world. Using this, we can state:

FUTURE VALUE For any x, t, t' where t is no later than t' , the credence function x should have at t Values the should-frame for x and t' .

This still faces some worries about forgetting. But these don't arise for the special case where $t = t'$, which is already interesting and potentially controversial:

VALUE For any x, t , the credence function x should have at t Values the should-frame for x and t .

That's because if we thought that the credence function x should have at t always assigns credence 1 to the proposition that it *is* the credence function x should have at t , Future Value would entail that credences should evolve by conditionalization.

4 Simple Trust

- Probability function π **Simply Trusts** frame $\mathcal{P}$ iff for every proposition $q \subseteq W$ and every $t \in [0, 1]$, $\pi(q \mid \mathcal{P}(q) \geq t) \geq t$.¹

Valuing entails Simple Trusting: for a given q and t , we can consider the decision problem $D_{q,t}$ whose two options are the constant function that maps every world to 0 ('no bet'), and the function that maps q -worlds to $1 - t$ and $\neg q$ -worlds to $-t$ ('bet on q at odds $t : 1 - t$ '). (Indeed, Simple Trusting is equivalent to the restriction of Valuing to the simplest possible non-trivial decision problems: decision problems with two options, one of which is constant, and the other of which has only two possible values.)

5 A challenge from Williamson's clock

Williamson's model for the clock case: $W = \{0, \dots, n - 1\}$; $\mathcal{P}(w) = \pi_0(\cdot \mid E(w))$, where π_0 is the probability function that gives equal probability to all worlds, and $E(w) =$

¹Here, $\mathcal{P}(q) \geq t := \{w : \mathcal{P}(w)(q) \geq t\}$.

$\{w-r, w-r+1, \dots, w, \dots, w+r-1, w+r\}$ for some “accessibility radius” $r > 0$. (Addition is mod n .)

Interpretation: $E(w)$ is your *total evidence* at w ; $\mathcal{P}(w)$ is your *evidential probability* function at w —the one that fits your total evidence.

Observation (Dorst): no probability function Simply Trusts $\mathcal{P}$. Take $r = 1$; then $\mathcal{P}(\{m-1, m+1\}) \geq 1/2 = \{m\}$. So if $\pi(\{m\}) > 0$,

$$\pi(\{m-1, m+1\} \mid \mathcal{P}(\{m-1, m+1\}) \leq 1/2) = 0 \not\geq 1/2$$

But this isn’t a terribly strong challenge to Value, since it is not intuitively plausible that you *should* have a credence function that assigns the same probability to all worlds within some interval and 0 to all worlds outside that interval. Credences that drop off gradually from the maximum seem much more reasonable.

- Someone who took very seriously the evidentialist slogan that ‘you should have the credences that fit your total evidence’ might feel forced to endorse Williamson’s model.² But (like Dorst) I’m not that big a fan of the ideology of evidence, so I don’t find this sort of argument very compelling.

6 Some Value-friendly models of Williamson’s clock

Here are some frames on $W = \{0, \dots, n-1\}$ are Valued by the flat prior, and also Value themselves (Dorst, p.c.):

- $\mathcal{P}(w)(\{w\}) = 1/2$ and $\mathcal{P}(w)(\{w-1\}) = \mathcal{P}(w)(\{w+1\}) = 1/4$.
- $\mathcal{P}(w)(\{w\}) = 1/3$, $\mathcal{P}(w)(\{w-1\}) = \mathcal{P}(w)(\{w+1\}) = 2/9$, $\mathcal{P}(w)(\{w-2\}) = \mathcal{P}(w)(\{w+2\}) = 1/9$.
- $\mathcal{P}(w)(\{w\}) = 3/10$, $\mathcal{P}(w)(\{w-1\}) = \mathcal{P}(w)(\{w+1\}) = 2/10$, $\mathcal{P}(w)(\{w-2\}) = \mathcal{P}(w)(\{w+2\}) = 1/10$, $\mathcal{P}(w)(\{v\}) = 1/10(n-5)$ when $|v-w| > 2$.

These seem more appealing than Williamson’s flat credences, since they capture the idea that the. But they involve some arbitrary-looking choices, it’s not obvious how sensitively their Value-friendliness depends on these choices. Here (finally) is where my little paper comes in.

7 Normal distribution frames can’t be Valued

Normal distributions (a.k.a. bell curves) arise all over the place in nature: when the states of two macroscopic systems are correlated but not perfectly correlated, typically we get

²One would need to appeal to the principle that if π fits E and π' fits E' , then $\pi(\cdot \mid E) = \pi'(\cdot \mid E)$ if both sides are defined.

an at least approximately normal distribution for the disparity between them (along any given dimension, for some choice of co-ordinates on that dimension). Plausibly, this will include the imperfect correlations that our imperfect sense-organs can induce between our credences and the external world. I think this makes it attractive to think that the best way for a creature with imperfect sense organs to be setup involves a disposition to end up, given flat priors, with normally-distributed credences about relevant external-world parameters.

The normal-distribution frame: $W = \mathbb{R}$, $\mathcal{F}$ is the algebra of Lebesgue-measurable sets, $\mathcal{P}(w)$ is the normal distribution with standard deviation 1 centred at w .³

Fact: no probability function Trusts the normal-distribution frame.

Proof: Pick small ε , e.g. 0.05 degrees, and consider the proposition q that the temperature is within ε of -1 or 1.

Of all the normal distributions with unit standard deviation, the one that maximizes the probability of q is the one centered at 0, which assigns it a probability of approximately 1.41ε .

By contrast, normal distributions centred near -1 and +1 assign q a probability of around 1.28ε .⁴

So, the only normal distributions that assign q a probability above 1.3ε are distributions whose centre is close enough to 0 that it is *not* within ε of -1 or +1.

So we have a failure of Simple Trust: any π that assigns positive probability to this range will be such that $\pi(q \mid \mathcal{P}(q) \geq 1.3\varepsilon) = 0 \not\geq 1.3\varepsilon$.

8 Convex frames can't be Valued

All that mattered about the normal distribution with s.d. 1 for the purposes of the previous section's argument is the fact that (a) its expected value is 0 and it takes its maximum value at 0, and (b) for some x , $f(0) + f(x) < 2f(x/2)$ and $f(0) + f(-x) < 2f(-x/2)$. Many many distributions work like this!

³To make it more realistic for cases like judging temperature, we could consider just some closed interval of real numbers; describing what happens to $\mathcal{P}$ near the cutoff will introduce some mathematical complexities but won't affect the key point. Alternatively, we could follow Williamson and consider $[0, 2\pi)$ (interpreted as clock-hand-angles). This will also complicate the math a bit since we'll need to switch to a 'wrapped' version of the normal distribution, but again that won't affect the key point.

⁴The probability density function of a normal distribution with unit standard distribution centred on T_1 is $y = e^{-(x-T_1)^2/2} / \sqrt{2\pi}$. So the probability of q in the distribution centred at T_1 is approximately $4\varepsilon e^{-(1/2)^2/2} / \sqrt{2\pi} \approx 1.41\varepsilon$, while its probability in the distribution centred at $T_1 - 1/2$ is $2\varepsilon(e^0 / \sqrt{2\pi} + e^{-1/2} / \sqrt{2\pi}) \approx 1.28\varepsilon$. The use of a small ε here is just to keep the calculations easy—the problem also arises for quite large values of ε .

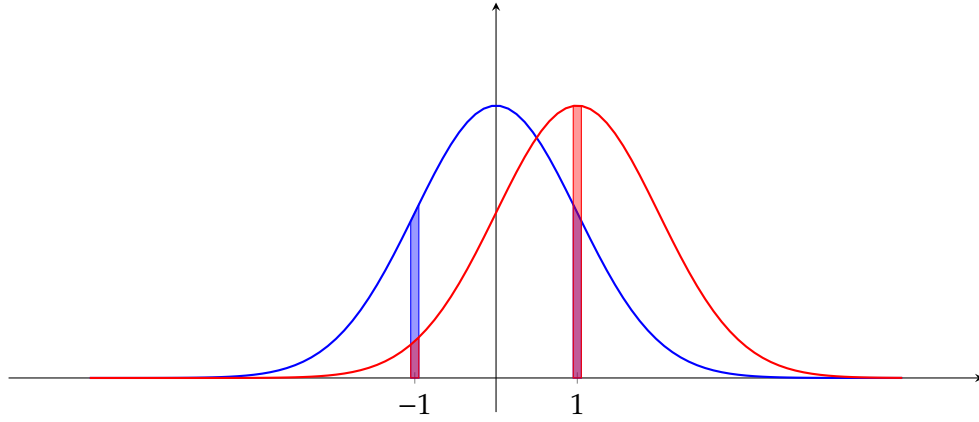


Figure 1: The blue bell curve depicts a credence function centred at T_1 ; the red one is centred at $T_1 + 1$. The shaded areas have width 2ε (here $\varepsilon = 0.05$) and are centred at -1 and 1 . It is not too hard to see (though a ruler might help) that the blue area is bigger than the red area.

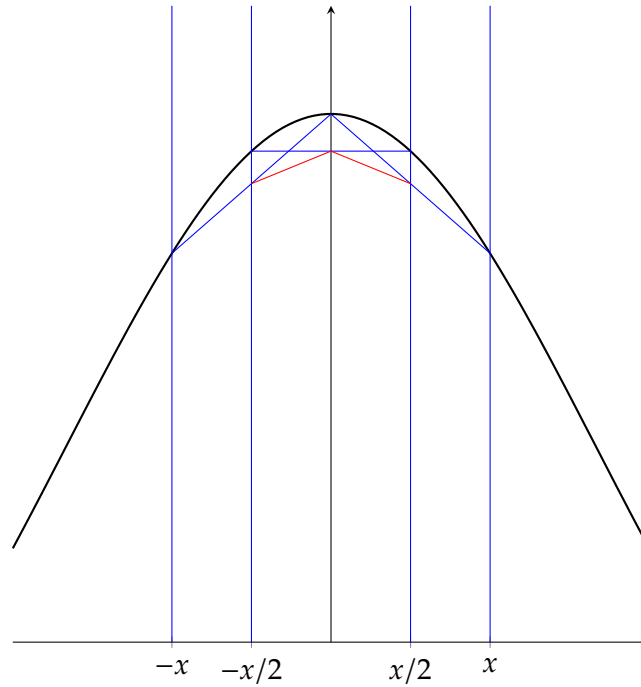


Figure 2: Convexity around 0

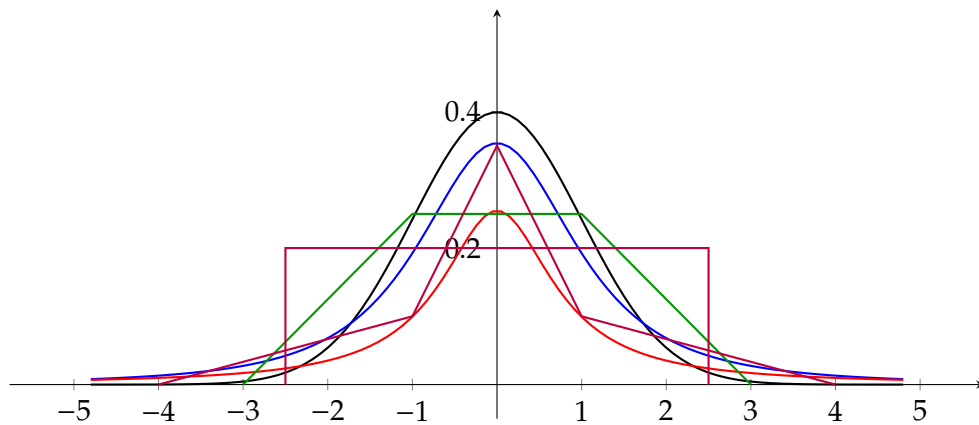


Figure 3: Probability densities convex around 0.

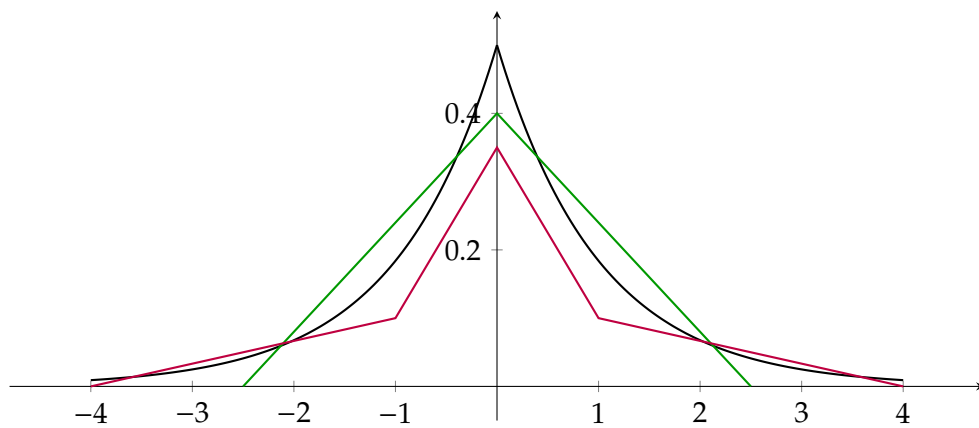


Figure 4: Probability densities centred on 0 but not convex around 0.

9 Do any real world cases work like this?

Does it ever really happen that there is some one-dimensional parameter such that *what our credences should be* about the value of that parameter supervenes (at least over some range of relevant worlds) on the value of that parameter?

You might think so, if you identify ‘what your credences should be’ with something like ‘what your credences would be if your sensory systems were functioning optimally’. For cases where your estimate is far off the truth seem like cases where you’re functioning non-optimally, so the best case scenario is when your estimate is exactly right.

On the other hand, this ‘should’ is supposed to have to do with ‘rationality’, and none of these failures involves anything we’d intuitively call irrationality. Someone might think that what your credences should be covaries with something causally intermediate between your credences and the worldly facts, such as your retinal state. Or they might even think (more weirdly!) that what your credences should be covaries with what your credences *are*, e.g. that in each world where your estimate (expected value) of the temperature is x , x also *should* be your estimate of the temperature.

10 The ‘one dimensional path’ strategy

11 Back to Valuing our future selves